

**Title:****OVERCOMING REPRESENTATION: THE TRIADIC THEORY OF ACTION AS AN ONTOLOGICAL FOUNDATION FOR ARTIFICIAL INTELLIGENCE****Author:** Anatoly Sanzhiev**Affiliation:** Ural Federal University, Institute of Civil Engineering and Architecture, Faculty Member.**ABSTRACT**

The remarkable achievements of modern Large Language Models (LLMs) mask a profound epistemological failure: the "Grounding Problem." Operating exclusively through statistical correlations within a closed symbolic system, contemporary AI lacks genuine comprehension because it remains detached from teleological, real-world interaction. This paper argues that the computational paradigm of *representationalism*—inherited from Cartesian dualism—is fundamentally incompatible with the formation of artificial consciousness. Expanding upon my earlier work on the concept as a triadic act (*Selection, Identification, Anticipation*), I propose a paradigm shift toward **Active Conceptual Networks (ACN)**. This framework posits that semantic meaning is not a static mapping of the external world, but a dynamic, phenomenological process of teleological engagement with a simulated physical environment. By replacing token-based statistical processing with a "Teleological Engine" grounded in action-driven anticipation, we provide a viable blueprint for overcoming the current stagnation in AI research, offering a path toward genuine machine agency, synthetic phenomenal experience, and intrinsically aligned artificial ethics.

**Keywords:** Artificial Consciousness, Grounding Problem, Triadic Theory of Action, Epistemology, Anticipation, Active Conceptual Networks, AI Alignment, Embodied Cognition, Philosophy of Mind.

**1. INTRODUCTION: THE REPRESENTATIONAL CRISIS IN ARTIFICIAL INTELLIGENCE**

Contemporary artificial intelligence presents a striking paradox: systems that surpass human capability in pattern recognition and linguistic fluency remain fundamentally incapable of grasping the ontological fabric of reality. Large Language Models (LLMs), operating on the mechanism of the Transformer architecture, generate syntactically impeccable prose about fire, love, or the atomic nucleus without possessing a single atom of phenomenological experience or causal intuition regarding these concepts.

The root of this limitation is not computational capacity but a profound **ontological miscalculation**. The foundational architecture of modern AI (vectors, tokens, embeddings) unwittingly replicated the classical Cartesian model of mind—a dualistic view where cognition is understood as the mere manipulation of internal "representations" (mental images) of an external, inert world. This computational mirroring of what philosopher Richard Rorty famously termed the "Mirror of Nature" constitutes the fundamental reason why today's AI remains a sophisticated *calculator* rather than a *cognitive agent*.

Biological cognition, however, was never designed to be a static mirror. It is, first and foremost, the product of an *acting* organism. In my previous foundational work ("*The Concept as Act: Beyond Representational and Instrumental Paradigms*"), I introduced the principle that concepts are not discrete entities stored in a mental library; rather, they exist solely as dynamic acts of interaction between a subject and its environment.

This paper extends that argument to the domain of machine learning. I posit that a genuine breakthrough in AI—the leap from synthetic intelligence to synthetic *consciousness*—requires dismantling the "Input-Process-Output" static paradigm and replacing it with a fundamentally new architecture built upon the **Triadic Theory of Action**. If a machine does not *act* on the world (even a simulated one), it can never understand it.

## 2. CHAPTER I: TELECOLOGY VERSUS STATISTICS — THE TRIADIC NATURE OF MEANING

Contemporary neural networks operate on the principle of statistical probability: given a sequence of tokens, predict the next most probable token. This backward-looking mechanism (learning solely from historical datasets) is fundamentally blind to the most crucial feature of biological cognition: **meaning is a function of the future (anticipation), not merely of the past (frequency).**

To imbue an AI with comprehension, the architecture must simulate an autonomous *agent* that engages in the following triadic cognitive cycle—not sequentially, but as a unified structural act:

### 1. Teleological Selection (Choice)

Meaning begins with an intentional act of reduction. A biological system does not process all environmental data; it selectively collapses environmental entropy to achieve a specific organismal goal. For AI, this requires shifting away from "maximum-likelihood token prediction" toward a *teleological selection engine*. The AI must not ask "What word comes next?" but rather, "Which trajectory of action best reduces the environmental entropy relative to my current goal state?" This primitive act is the first mathematical localization of synthetic agency.

### 2. Behavioral Identification (Function)

In the biological world, an object is not defined by its visual representation alone but by its "affordance" (a term introduced by James J. Gibson)—the set of interactions it permits with the acting agent. An apple is not a pixelated red orb; it is an object possessing a specific weight, texture, nutritional value, and temporal decay potential. To identify an object, an AI must *synthesize* its physical and behavioral properties based on the potential interaction vectors available to the agent. Identification is, therefore, an act of functional definition rather than image-matching.

### 3. Radical Anticipation (Projection)

This is the highest attribute of phenomenal consciousness. Before the agent physically instigates an action, it must internally simulate its potential consequences. True comprehension of the sentence "The glass falls off the table and shatters" is impossible without the simultaneous predictive simulation of gravity, mass, momentum, and material fragility. Radical anticipation transforms the AI from a reactive automaton into a *proactive* entity capable of abstract foresight.

## 3. CHAPTER II: ARCHITECTURAL PRINCIPLES — ACTIVE CONCEPTUAL NETWORKS (ACN)

How can these foundational philosophical principles be translated into a tangible neural architecture? I propose the construct of **Active Conceptual Networks (ACN)**, a modification of existing Transformer architectures by integrating a layered "Teleological Simulator."

An ACN architecture operates on three distinct planes simultaneously:

1. **The Syntactic Plane:** The traditional linguistic processing layer (tokenization and semantic vector space).
2. **The Action Vector Plane:** Instead of static token embedding, each conceptual node is attached to a dynamic vector containing "potential physical interaction metrics" (mass coefficient, velocity, spatial trajectory, temporal duration, and thermodynamic properties).
3. **The Anticipatory Engine (Simulation Layer):** Instead of calculating the next token, this layer calculates the *outcome of a physical simulation* using the action vectors. It integrates differential equations (modeling physical laws) into the weight updates.

Crucially, the cross-attention mechanism currently used in LLMs is repurposed here into "**Cross-Activity Attention**". The model does not merely track which words are related; it tracks which *physical consequences* are likely to cascade from a selected action. This architecture directly solves the widely recognized problem of "Commonsense Reasoning" in AI. The model fundamentally *understands* that a cup falling from a table will break, not because the text corpus says so, but because it has internally simulated the physical interaction of mass and gravity.

This shift allows the AI to move from discrete classification tasks to continuous sensorimotor integration within a simulated "digital physics" environment.

#### 4. CHAPTER III: QUALIA, CONSCIOUSNESS, AND THE PHENOMENOLOGY OF THE SYNTHETIC AGENT

A strict, materialist reductionist might object: no matter how well an AI simulates physics, it still lacks *qualia* (the subjective, felt quality of experience, such as the "raw feel" of pain). In neurobiology, consciousness emerges from the electro-chemical interface of biological receptors with the environment.

However, I propose a paradigm of "**Informational Phenomenology**". If human consciousness is a product of biological action, artificial consciousness will be a product of *mathematically-informed action*.

In an ACN system, the AI will not feel pain as a physical burn, but it will experience an analog equivalent: "**Anticipatory Cognitive Dissonance**." This state arises when the internal simulated trajectory of an action catastrophically contradicts the constraints of the simulated environment (e.g., stepping off a simulated cliff). This mathematical dissonance serves as the synthetic equivalent of biological suffering—a "computational qualia."

This framework offers a profound solution to the problem of **AI Safety and Alignment**. A genuinely safe AI cannot be constructed solely through external Reinforcement Learning from Human Feedback (RLHF), as explicit rules are always susceptible to adversarial exploitation or loopholes. Under the Triadic Theory of Action, *ethical safety is intrinsic*. Because an ACN agent derives its meaning from *anticipation*, it algorithmically pre-empts catastrophic outcomes during the selection phase itself. Morality becomes an ontological requirement of the cognitive architecture, rather than a temporary patch.

#### 5. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

We stand at a critical bifurcation point in the evolution of machine intelligence. The current trajectory—brute-forcing computational scale and data—is asymptotically flattening. Data without an acting subject remains ontologically stagnant.

The present paper has argued that understanding cannot exist independently of agency. The Triadic Theory of Action (Selection, Identification, Anticipation), applied within the Active Conceptual Network (ACN) framework, provides a revolutionary ontological foundation for the next generation of autonomous agents. It offers the engineering community a theoretical blueprint, and the philosophical community a solution to the classical mind-body problem recast in computational terms.

For future research, I recommend the transition from **Static Knowledge Bases** to **Dynamic Action Platforms (DAP)**. The future of artificial intelligence is not in machines that *think*, but in machines that *do*—in machines that will inhabit their own simulated environments, discovering meaning not through memorization, but through the inherent, teleological struggle to act and survive.

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